

YANJUN YANG

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EDUCATION

The University of Edinburgh

PhD Candidate in Engineering, IMNS

Sept, 2022 - present

Edinburgh, Scotland

- Supervisors: Dr Alex Serb, Prof Themis Prodromakis
- Research theme: Hardware architectures for accelerating graph databases in hybrid neuro-symbolic AI
- Fully funded by a University of Edinburgh School of Engineering Studentship

Nanyang Technological University

MSc (Electronics)

Aug, 2021 - Aug, 2022

Singapore

- Core units taken: Digital Integrated Circuit Design, Electromagnetic Compability Design, Genetic Algorithms and Machine Learning, Advanced Wafer Processing, Integrated Circuit Packaging

Tongji University

BEng. in Electronic Science and Technology

Sept, 2017 - July, 2021

Shanghai, China

- Core units taken: Design and Analysis of Digital Integrated Circuits, Computer Architecture, Principles for Design of Integrated Circuit Chips, Embedded Systems

CURRENT PROJECTS

ASOCA3

The University of Edinburgh

Jan, 2024 - present

PhD Sub-project

- Developing and iterating a high-performance FPGA-based graph database accelerator SoC
- Leading the system architecture and processing-core microarchitecture design, co-developing the graph database model, and defining the instruction-set specification
- Supporting the development of an ASOCA3-based vector database accelerator and a DRAM-based dually-addressable memory design

ASOCA Infrastructure Development Environment

The University of Edinburgh

Nov, 2025 - present

PhD Sub-project

- Developing infrastructure tooling, design-flow automation, and a tailored UVM system-verification environment for the ASOCA family and related projects
- Building workload-generation and performance-benchmarking tools using Neo4j and Memgraph
- Contributing to the integration and verification of NoC and PCIe subsystems

EMPLOYMENT

Research Assistant

The University of Edinburgh

March, 2026 - Present

Edinburgh, Scotland

- Contributing to the development of the ATLAS subsystem within the wider ADIA2 project
- Supporting technical direction, system development, and commercialisation planning
- Providing technical support and consultancy as part of the project team

Teaching Assistant

The University of Edinburgh

January, 2024 - February, 2026

Edinburgh, Scotland

- Tutoring, demonstrating and marking for multiple undergraduate courses.

PUBLICATIONS

Views: A Hardware-friendly Graph Database Architecture For Storing Semantic Information August, 2025

Preprint [\[arXiv\]](#)

- Authors: **Yanjun Yang**, Adrian Wheeldon, Yihan Pan, and Alex Serb

A Resource-efficient Dually-addressable Memory Architecture on FPGA May, 2025
ISCAS 2025 London, United Kingdom

- Authors: **Yanjun Yang**, Christos Giotis, Themis Prodromakis and Alex Serb

ASOCA2: A Modular Graph Database Accelerator September, 2024
ECML/PKDD 2024 Vilnius, Lithuania

- Authors: Yihan Pan, Adrian Wheeldon, **Yanjun Yang**, Khaled Humood, Mohammed Mughal, Graham Reynolds, Themis Prodromakis, and Alexantrou Serb

A Single-layer Wideband Microwave Absorber with Reactive Screen, and A Novel Design of Microwave Absorber for Reduction of Radar Cross Section July, 2020
AP-S/URSI 2020 Montréal, Canada

- Carried out antenna simulation under HFSS and experimental data analyses

RESEARCH EXPERIENCE

Saliency detector Feb, 2024 - July, 2024
The University of Edinburgh, STMicro PhD Sub-project

- Developed a hybrid neuro-symbolic prototype for semiconductor defect detection and classification
- Produced a results-analysis and development report

ASOCA2 Nov, 2022 - Dec, 2023
The University of Edinburgh PhD Sub-project

- Developed a scalable graph database accelerator prototype using a memristive dually-addressable memory array
- Led RTL development, functional verification, and top-level integration of the core digital modules
- Contributed to synthesis and tape-out delivery, followed by chip bring-up, functional testing, and evaluation

ASOCat Sept, 2022 - Jan, 2024
The University of Edinburgh PhD Sub-project

- Investigated a Copycat-based cognitive model compatible with an associative-memory chip
- Designed and implemented a graph database storage method and its software-hardware interface
- Explored hardware implementations of procedural memory and lambda-calculus models

ReMap: a Mitchell-based logarithmic conversion circuit Aug, 2021 - July, 2022
Nanyang Technological University MSc Dissertation Project

- Optimised a Mitchell-based binary logarithmic approximation method
- Implemented and evaluated corresponding integrated circuits

Design of a hierarchical memory management mechanism

Tongji University

Aug, 2021 - Jan, 2022

Part-time Internship

- Led a hierarchical SRAM-flash interface design with page replacement algorithm
- Applied the design on an automobile-orientated MCU

CoNM: Core of Normal Microarchitecture

Tongji University

Mar, 2021 - Jun, 2021

Graduation Design

- Designed a four-stage RV32I CPU core with static branch prediction in Verilog
- Implemented onto PYNQ-Z1 FPGA board for a successful verification

PROFESSIONAL SKILLS

Hardware Design

- Proficient in SystemVerilog/Verilog, SVA, UVM, and functional verification
- Experienced with FPGA and digital ASIC design flows
- Completed Europractice training in advanced-node digital IC design, including high-performance and low-power design techniques

EDA and Engineering Software

- Synopsys, Vivado, Cadence, ModelSim, MATLAB, HFSS, ISE, and Keil

Software and Scripting

- Python, Git, (Neo)vim, Bash, Tcl, C/C++, VHDL, and Chisel

Architecture and Research Areas

- RISC-V, graph databases, arithmetic circuits, hybrid neuro-symbolic AI, and memory systems

Languages

Native in Mandarin and New Xiang, proficient in English

- IELTS 8.0/9.0 (2020), equivalent to CEFR level C1
- Elementary proficiency of French

SEMINARS & TALKS

IMNS Seminar

Institute for Integrated Micro and Nano Systems

November, 2024

Edinburgh, Scotland

- Title: Graph Database Acceleration in Digital Hardware

PGR Conference

The University of Edinburgh

April, 2024

Edinburgh, Scotland

- Title: Symbolic Processing Systems Based on Associative Memory

TEACHING

The University of Edinburgh

1st Semester, 2024

- Demonstrator - Digital Systems Design 2 (ELEE08015)
- Marker - Software and Embedded Systems Laboratory 2 (ELEE08022)

The University of Edinburgh

2nd Semester, 2024

- Demonstrator & Marker - Digital Systems Laboratory 3 (ELEE09018)

The University of Edinburgh

1st Semester, 2025

- Tutor & Marker - Digital Systems Design 2 (ELEE08015)
- Marker - Software and Embedded Systems Laboratory 2 (ELEE08022)

HONOURS AND AWARDS

The University of Edinburgh

Mar, 2022

School of Engineering Studentship

- Full studentship for full-time PhD study

Tongji University

1st Semester, 2017

Advanced Summer Internship Individual

- Awarded for the great performance during summer internship

Tongji University

1st Semester, 2019

Outstanding Student Cadre

- Awarded as the minister of Rights and Welfare Department in Students' Union

INTERESTS

Scotch whisky enthusiast.

Half geek, half bartender.

Neovim sponsor and WSL2 user.

Reads Kafka, Hai Zi, and Edgar Allan Poe.

Fan of Jay Chou, Green Day, and Aska Yang.

Competitive Minesweeper player. (Best time: 107.269s)